

Collective Roadmapping and Participatory Futures: How Teams Plan Together

Executive Summary

Planning is too important to be left to executives alone. Under Active Inference, **collective planning** harnesses the distributed knowledge and diverse models of the entire team (or organization) to construct richer scenario spaces and more robust strategies. This module covers participatory planning processes, collective foresight methods, the concept of emergent strategy (Mintzberg), and how to design planning processes that balance top-down direction with bottom-up intelligence.

Learning Objectives

1. Explain **collective planning** as distributed policy evaluation across many agents
 2. Understand **emergent strategy** (Mintzberg) — how strategy arises from bottom-up action, not just top-down planning
 3. Apply **participatory foresight** methods (Delphi, futures wheels, backcasting)
 4. Design planning processes that **integrate top-down direction with bottom-up intelligence**
 5. Evaluate when collective planning outperforms expert planning
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Key Concepts

1. Emergent Strategy (Mintzberg)

Henry Mintzberg's key insight: much of an organization's actual strategy is **emergent** — it arises from patterns of local decisions, not from the deliberate top-down plan:

- **Deliberate strategy**: What was intended (the strategic plan)
- **Realized strategy**: What actually happened (the pattern of actions)

- **Emergent strategy:** The gap between the two — strategy that arose from bottom-up decisions

Active Inference framing: Emergent strategy is the aggregate result of distributed policy selection by many local agents, each minimizing their own expected free energy. If these agents are well-aligned, emergence produces coherent strategy. If misaligned, emergence produces strategic drift.

2. Participatory Foresight Methods

Method	Process	Strength
Delphi method	Multiple rounds of anonymous expert input, with feedback	Aggregates diverse expertise without social pressure
Futures wheels	Map cascading consequences of a change	Reveals second- and third-order effects
Backcasting	Start with a desired future, work backward to identify required actions	Overcomes anchoring on present conditions
Planning cells	Random citizens/employees deliberate on complex issues	Incorporates diverse perspectives
Scenario workshop	Groups co-create and evaluate future scenarios	Builds shared ownership of strategic direction

3. Bottom-Up Intelligence in Planning

Frontline employees often have the richest sensory data about customer needs, operational realities, and emerging problems. Effective collective planning creates channels for this intelligence to inform strategy:

Case Study — Toyota's A3 Process: Toyota's A3 problem-solving process empowers frontline workers to propose and implement improvements. Each A3 is a one-page document that captures the problem, analysis, and proposed solution — a structured mechanism for

bottom-up intelligence to shape organizational action. This process ensures that planning is not monopolized by executives but continuously informed by those closest to the work.

4. Designing Collective Planning Processes

Principles for effective collective planning:

1. **Diverse inputs:** Include voices from multiple levels and functions
 2. **Structured aggregation:** Use formal methods to combine diverse views (not just "loudest voice wins")
 3. **Iterative refinement:** Multiple rounds of input and feedback
 4. **Clear decision rights:** Participation in planning $\neq$ decision authority — clarify who decides
 5. **Feedback loops:** Track how bottom-up inputs influenced the final plan
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Cross-References

- For strategic planning and resilience, see [Organizational Systems: Planning](#)
 - For risk management and scenario planning, see [Strategic Modeling: Planning](#)
 - For digital transformation roadmaps, see [Digital Transformation: Planning](#)
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Summary

Concept	Team/Collective Meaning
Collective planning	Distributed policy evaluation leveraging team knowledge
Emergent strategy	Strategy arising from bottom-up local decisions, not just top-down plans
Participatory foresight	Methods (Delphi, backcasting) for harnessing collective future-thinking
Bottom-up intelligence	Frontline insights informing organizational planning
Planning process design	Structuring collective input for effective strategy development

References

- Mintzberg, H. (1987). The strategy concept I: Five Ps for strategy. *California Management Review*, 30(1), 11–24.
- Rowe, G., & Wright, G. (1999). The Delphi technique as a forecasting tool. *International Journal of Forecasting*, 15(4), 353–375.